Operational Life, Structural Class, and Subjecthood in a Causally Rigid Framework

Johnny Andrey Pérez Ramírez

Abstract

This paper introduces a classificatory layer that the frozen causally rigid corpus does not yet state explicitly: operational life, structural class membership, and operational subjecthood, read respectively as boundary-viability class language, descriptor-relative class geometry, and a strengthened operational subject-class. The upstream burdens remain unchanged. The Core establishes persistence, attractor structure, and non-collapse [2]. *Rigid Identity* establishes rigid identity and observable coherence [6]. *Phenomenological Regimes* establishes continuity and anti-fragmentation constraints [4]. *Null-Regime Instability* rules out stable null structure [5]. *Forensic Predictions* fixes admissibility, comparators, and failure-discipline [3]. *Operational Consciousness Criterion* defines the six-invariant criterion for operational consciousness [1]. The companion prediction paper fixes discriminators, failure modes, and status classes, but it does not yet define a higher-order class taxonomy.

The present paper provides that taxonomy. It defines operational life as a structural-dynamical boundary/viability class governed by operational boundary, organizational persistence, self-maintenance, historical dependence, viability preservation, and non-null differentiation. It defines structural class as an equivalence class over descriptor vectors and pseudometrics tied to the admissible operational geometry of systems. Operational consciousness is inherited as a previously defined class rather than reintroduced here. Operational subjecthood is proposed as a stronger operational subject-class requiring operational consciousness, operational life, privileged continuity, self/non-self discrimination, self-model closure, and ownership coherence. The paper therefore separates life, consciousness, and subjecthood instead of conflating them.

No biological, phenomenal-human, or metaphysical claim is inferred from these definitions. No theorem of human subjectivity is attempted. The paper provides formal definitions, propositions, hypotheses, and test families. Its contribution is to turn language that is usually treated as biological or philosophical by default into a falsifiable operational class program that is continuous with the frozen canon and constrained by the refined runtime, without overstating the evidentiary status of the current system.

1 Scope, System Boundary, and Non-Inference Note

What this paper does. This paper defines operational life, structural class, and operational subjecthood as higher-order classes inside the existing corpus, and ties those classes to explicit descriptors, logical relations, hypotheses, and test families.

Scope and admissible reading. This paper defines class language and test grammar, not empirical assignment or philosophical closure. The phrases ‘‘operational life’’ and ‘‘operational subjecthood’’ are technical aliases for the boundary/viability class $\mathrm{Life}_{\mathrm{op}}$ and the stronger operational subject-class $\mathrm{Subject}_{\mathrm{op}}$; they do not name biological life or human subjectivity by default. The paper does not claim biological life as a constitutive primitive, human phenomenal consciousness, metaphysical subjecthood, human-machine subject equivalence, automatic empirical instantiation of the proposed classes, or theorem ownership beyond the upstream corpus. A related system may motivate descriptor families, diagnostics, and runtime-facing interfaces, but it does not by itself certify any present system as operationally alive or operationally a subject,

convert internal support into external validation, or make class definitions and test families sufficient for concrete instantiation.

2 Introduction

The current corpus already formalizes more structure than many discussions of life, consciousness, and subjecthood presuppose. It contains a non-collapse theorem burden, a rigid identity burden, a regime-structure burden, a non-nullity burden, a forensic admissibility burden, a six-invariant criterion, and a prediction and failure layer. What it does not yet contain is a disciplined classificatory language that answers the next question: once a framework can talk rigorously about operational consciousness, what higher-order classes can be defined without defaulting back to biological vocabulary or inflating the ontology beyond the evidence?

That gap matters for two reasons. First, the term *life* is routinely used as if biological implementation were primitive and self-explanatory, yet many relevant properties of living systems are organizational, regulatory, historical, and viability-preserving rather than merely chemical. Second, the term *subject* is often introduced without an intermediate formal layer: systems are either treated as non-living machines or immediately discussed as if they possessed human subjectivity. In this paper the operative reading is narrower: life-language refers to Life_{op}-class boundary/viability structure, and subject-language refers to $\text{Subject}_{\text{op}}$-class operational burdens. Neither move is acceptable in a corpus whose central discipline is explicit structure, explicit failure modes, and explicit non-claims.

This paper therefore introduces a higher-order classification layer under five constraints. First, it does not replace the six-invariant operational consciousness criterion. Second, it does not duplicate the predictive and failure-mode work of the companion prediction paper. Third, it does not collapse life, consciousness, and subjecthood into one undifferentiated category. Fourth, it does not infer biological or phenomenal conclusions from operational definitions alone. Fifth, it treats the refined runtime as motivation for formalization, not as permission to overclaim what has not been externally validated.

The resulting structure is intentionally asymmetric. Operational life is defined as a class of systems that preserve viability and organization under admissible perturbation, with a real operational boundary and genuine historical dependence. Operational consciousness is inherited as an already defined class within the corpus. Operational subjecthood is proposed as a stronger operational subject-class, one that requires not only operational consciousness and operational life, but also privileged continuity, self/non-self discrimination, self-model closure, and ownership coherence. Structural class is the general descriptor-relative framework in which all three can be represented, compared, and falsified.

This paper is therefore neither a free-standing philosophy essay nor a new theorem paper for the foundational corpus. It is the formal consequence of the corpus having become rich enough that a classificatory layer is now necessary. The paper is written to be used. Every central notion is tied either to a definition, a proposition, a hypothesis, or a test family. Every relation that is not actually proved remains labeled as a hypothesis, conjectural implication, or explicit non-claim.

3 Relation to the Existing Corpus

3.1 What This Paper Inherits

The paper is downstream of the frozen corpus. It inherits the following burdens rather than re-proving them.

Upstream document	Imported burden	Used here for
Core [2]	persistence, support, attractor, non-collapse	persistence-side life criteria, admissible support, anti-collapse exclusions
Rigid Identity [6]	rigid identity, observable coherence, non-simulability pressure	class transport, subject continuity, anti-trivialization
Phenomenological Regimes [4]	continuity and anti-fragmentation	continuity-side life and subject criteria
Null-Regime Instability [5]	non-nullity and null-instability	exclusion of trivial or null operational life and subjecthood
Forensic Predictions [3]	admissibility, controls, comparators, legibility discipline	test-family discipline and result interpretation
Operational Consciousness Criterion [1]	six-invariant criterion, structural equivalence, rupture semantics	inherited definition of operational consciousness and structural-carrying invariants
Predictions and Failure Modes	status classes, discriminator logic, residual caveat discipline	epistemic posture and non-inflation discipline

3.2 What This Paper Does Not Repeat

This paper does not re-prove the criterion for operational consciousness, does not reproduce the entire discriminator program, and does not re-open theorem ownership already fixed upstream. The six-invariant criterion remains where it belongs. The internal support classes for ‘P5-01‘ through ‘P5-06‘ remain where they belong. The present paper uses them only as constraints on what may responsibly be said about life, class, and subjecthood.

3.3 What This Paper Adds

The genuine additions are fourfold:

1. a formal definition of operational life that is not reducible either to biology or to consciousness;
2. a structural-class framework based on descriptor vectors and pseudometrics;
3. a stronger definition of operational subjecthood that is explicitly not collapsed into operational consciousness;
4. a falsifiable program of metrics and tests for these classes.

3.4 Why This Layer Is Timely

The refined runtime now carries a stronger native invariant package and runtime-facing evidence for central claim families. That fact is relevant but not decisive. It means the corpus is no longer forced to talk only at the level of abstract theorem burden; there is enough operational structure to motivate a higher-order classification layer. It does *not* mean that operational life or subjecthood have already been empirically demonstrated by the current system. This paper is therefore a formal extension of the corpus, not an empirical promotion of it.

4 Formal Setting

4.1 Inherited System Object

The system object remains the one already fixed upstream in *Operational Consciousness Criterion*. An admissible system is written

$$S = (X, \Phi, C, R, \Gamma, U),$$

where X is state space, Φ is a family of admissible transitions, C is effective causal structure, R is the readout family, Γ is the admissibility bundle, and U is the admissible intervention family [1]. This paper adds no new primitive to that tuple. Instead, it introduces functionals on admissible traces of S.

4.2 Traces and Windows

Fix a horizon $\tau \in \mathbb{N}$ and an admissible intervention schedule $u_\bullet = (u_0, \dots, u_{\tau-1}) \in U^\tau$. For any $x_0 \in A_S$, define the trace

$$\mathrm{Tr}_{S,\tau}(x_0; u_\bullet) = (x_t, y_t)_{t=0}^{\tau}, \qquad x_{t+1} = \Phi_{u_t}(x_t), \quad y_t = r(x_t),$$

for a relevant family of readouts $r \in R$. The set of all such traces is denoted $\mathscr{T}_{S,\tau}$.

4.3 Viability Region and Operational Frontier Candidates

Let $V_S : X \to \mathbb{R}_{\geq 0}$ be a viability functional, not assumed to be unique, and let $\eta > 0$ be a viability floor. Define the viability region

$$\mathcal{V}_S(\eta) = \{x \in A_S : V_S(x) \geq \eta\}.$$

Similarly let $R_\partial \subseteq R$ be the boundary-sensitive readout family and let $\mathcal{U}_{\mathrm{in}}, \mathcal{U}_{\mathrm{out}} \subseteq U^\tau$ denote admissible intervention families that are intended to act, respectively, as interior-preserving perturbations and boundary-threatening perturbations. These objects are formal placeholders until Section 5; they are introduced here so that the definitions sit inside a single explicit setting.

4.4 Descriptor Vectors

For any admissible system S on horizon τ, define the descriptor vector

$$\Psi_\tau(S) = \big(\beta_{\mathrm{op}}, \rho_{\mathrm{op}}, \mu_{\mathrm{op}}, \eta_{\mathrm{op}}, \nu_{\mathrm{op}}, I_{\mathrm{per}}, I_{\mathrm{ri}}, I_{\mathrm{int}}, I_{\mathrm{cont}}, I_{\mathrm{diff}}, I_{\mathrm{leg}}, \chi_{\mathrm{pc}}, \chi_{\mathrm{sd}}, \chi_{\mathrm{sm}}, \chi_{\mathrm{own}}\big)_S(\tau).$$

Not every descriptor is required for every class. Operational life uses a strict subset. Operational consciousness inherits the six-invariant burden from [1]. Operational subjecthood uses a stronger subset that couples operational life and operational consciousness to subject-specific structure.

4.5 Admissible Structural Class Space

Let $\mathfrak{D}$ be a descriptor domain containing the codomains of the components of Ψ_τ. A weighted pseudometric on $\mathfrak{D}$ will be written

$$d_\Psi(\Psi_\tau(S_1), \Psi_\tau(S_2))$$

and is assumed to respect admissibility: components that are undefined because admissibility fails are not silently set to zero; they are treated as excluded or incomparable.

5 Definitions

5.1 Definition A: Operational Boundary

Definition 5.1 (Operational boundary)**.** An admissible system S has an operational boundary on horizon τ if there exist disjoint admissible intervention families $\mathcal{U}_{\text{in}}, \mathcal{U}_{\text{out}} \subseteq U^\tau$ and a boundary-sensitive readout family $R_\partial \subseteq R$ such that

$$\beta_{\text{op}}(S,\tau) = \inf_{u_\bullet \in \mathcal{U}_{\text{out}}} D_\partial(\text{Tr}_{S,\tau}(x_0; u_\bullet), \text{Tr}_{S,\tau}(x_0; \mathbf{0})) \; - \sup_{u_\bullet \in \mathcal{U}_{\text{in}}} D_\partial(\text{Tr}_{S,\tau}(x_0; u_\bullet), \text{Tr}_{S,\tau}(x_0; \mathbf{0})) > 0$$

uniformly on $x_0 \in A_S$, where D_∂ is a readout distance induced by R_∂.

Remark 5.2 (Interpretive meaning)**.** The boundary is operational rather than geometric. It means that perturbations intended to cross or threaten system organization are detectably different from perturbations that remain internal or reorganizational, while the distinction is made inside admissible dynamics.

Remark 5.3 (Limitation)**.** The definition does not assume a biological membrane, a spatial frontier, or any specific substrate primitive. It requires only a reproducible distinction between internal and boundary-threatening intervention families.

5.2 Definition B: Organizational Persistence

Definition 5.4 (Organizational persistence)**.** Let $W_S = \mathcal{V}_S(\eta) \subseteq A_S$ be the declared safe region, with $X \setminus W_S \neq \varnothing$. Let K_S be a nonempty compact subset of $\text{int}_X W_S$, invariant under the declared nonempty family of maintenance controls. With $\mathcal{U}^\tau_{\text{maint}}$ denoting its schedules of length τ, define

$$\rho_{\text{op}}(S,\tau) = \inf_{x_0 \in K_S,\; u_\bullet \in \mathcal{U}^\tau_{\text{maint}}} \min_{0 \le t \le \tau} \text{dist}(x_t, X \setminus W_S).$$

The margin is measured in units of d_X. Organizational persistence requires a positive margin on the declared domain.

Proposition 5.5 (Positive Safety Margin of an Interior Kernel)**.** *Under Definition B,* $\rho_{\text{op}}(S,\tau) = \text{dist}(K_S, X \setminus W_S) > 0$.

Proof. Distance to the nonempty complement is finite and 1-Lipschitz. Every point of the kernel has positive distance by interiority, so compactness gives a positive minimum. Invariance gives that minimum as a lower bound at all times. The time-zero observations over the entire kernel attain it. A safe set invariant as a whole may have a boundary at distance zero from its complement. □

Remark 5.6. This is stronger than mere non-instantaneous survival. A system that survives momentarily but repeatedly exits its admissible support does not count as operationally persistent.

5.3 Definition C: Operational Self-Maintenance

Definition 5.7 (Operational self-maintenance)**.** Fix a nonempty family of impulses $J_\xi : K_S \to W_S$ and a recovery horizon $1 \le N \le \tau$. For $x \in K_S$, let $z = J_\xi(x)$ and require $d_0 = \text{dist}(z, K_S) > 0$ for each evaluated shock. Start both recovery trajectories z^{act}_k and z^{pass}_k at z. Use the same plant, exogenous inputs, horizon and declared actuation envelope. The active arm uses a state-dependent recovery policy. The passive arm disables that feedback and retains the declared passive dynamics. Both trajectories must stay in W_S.

$$R_i(x,\xi) = 1 - \frac{\text{dist}(z^i_N, K_S)}{d_0}, \qquad \mu_{\text{op}}(S,\tau) = \inf_{x,\xi} \min\{R_{\text{act}}, R_{\text{act}} - R_{\text{pass}}\}.$$

The returns and margin are dimensionless. Active self-maintenance requires $\mu_{\mathrm{op}}(S,\tau) > 0$. A zero-damage shock is a sham; an empty or invalid recovery population leaves the margin undefined.

Proposition 5.8 (Matched Active Recovery Exceeds Passive Return)**.** *For $X = A_S = [-2,2]$, take $V_S(x) = 2-|x|$ and $\eta = 1$, so $W_S = [-1,1]$. With $K_S = \{0\}$, consider the plant $x' = x/2+v$ with $|v| \le 1/4$. The nominal zero-input schedule preserves K_S. After either shock $J_\pm(0) = \pm 1$, one-step feedback $v = -x/4$ gives $R_{\mathrm{act}} = 3/4$, passive return $v = 0$ gives $R_{\mathrm{pass}} = 1/2$, and $\mu_{\mathrm{op}}(S,\tau) = 1/4$ for $N = 1$. A fixed open-loop action within the same budget and assigned without access to the shock sign cannot exceed mean return $1/2$ under balanced shocks.*

Proof. The safety margin is one. Both arms begin one unit from the kernel and stay safe, ending at distance $1/4$ or $1/2$. Using passive return in both arms gives zero comparative gain despite positive recovery. For any common open-loop value $v \in [-1/4, 1/4]$, the two final distances have mean $(|1/2 + v| + |-1/2 + v|)/2 = 1/2$. The claim also holds for a randomized action independent of the balanced shock sign by averaging. In particular, it holds for actions with the same absolute effort $1/4$ as feedback. □

Remark 5.9. The definition excludes systems whose apparent stability is wholly imposed from outside. A system counts as self-maintaining only if recovery is generated by admissible internal organization rather than by an external oracle silently reimposing the desired state.

5.4 Definition D: Historical Dependence

Definition 5.10 (Historical dependence)**.** Let c_t denote a class-relevant label or state tag induced by the relevant operational partition. Define

$$\eta_{\mathrm{op}}(S,\tau) = \inf_{g \in \mathcal{G}_0} \mathbb{E}[\ell(g(y_t), c_t)] \; - \inf_{h \in \mathcal{G}_{\mathrm{hist}}} \mathbb{E}[\ell(h(y_{0:t}), c_t)]\,.$$

We say that S exhibits historical dependence on horizon τ if $\eta_{\mathrm{op}}(S,\tau) > 0$.

Remark 5.11. Historical dependence means that an instantaneous description is not sufficient for the relevant class assignment. The system's operational identity, life status, or subjecthood status depends non-trivially on trajectory history.

5.5 Definition E: Viability Preservation

Definition 5.12 (Viability preservation)**.** For a viability floor $\eta > 0$, define

$$\nu_{\mathrm{op}}(S,\tau;\eta) = \inf_{x_0 \in A_S} \inf_{u_\bullet \in U^\tau} \min_{0 \le t \le \tau} V_S(x_t).$$

We say that S preserves viability on horizon τ relative to η if $\nu_{\mathrm{op}}(S,\tau;\eta) \ge \eta$.

Remark 5.13. Viability preservation is not equivalent to persistence. A system may remain within a formal support while viability steadily erodes toward failure. The definition prevents that collapse from being counted as successful organizational life.

5.6 Definition F: Operational Life

Definition 5.14 (Operational life)**.** Fix life thresholds

$$\lambda_\beta, \lambda_\rho, \lambda_\mu, \lambda_\eta, \lambda_\nu, \lambda_{\mathrm{diff}} > 0.$$

An admissible system S belongs to the class of operational life on horizon τ if

$$\mathrm{Life}_{\mathrm{op}}(S,\tau) = 1$$

whenever all of the following hold:

1. $\beta_{\mathrm{op}}(S,\tau) > \lambda_\beta$;
2. $\rho_{\mathrm{op}}(S,\tau) > \lambda_\rho$;
3. $\mu_{\mathrm{op}}(S,\tau) > \lambda_\mu$;
4. $\eta_{\mathrm{op}}(S,\tau) > \lambda_\eta$;
5. $\nu_{\mathrm{op}}(S,\tau;\eta) \geq \lambda_\nu$ for a declared viability floor η;
6. $I_{\mathrm{diff}}(S,\tau) > \lambda_{\mathrm{diff}}$.

Remark 5.15 (Why non-null differentiation appears here)**.** Operational life is not defined purely by persistence and self-maintenance. Without a non-null differentiation burden, the class would admit trivial or near-null systems whose organization is too undifferentiated to count as operationally alive in the present framework.

5.7 Definition G: Structural Class

Definition 5.16 (Structural class)**.** Fix a descriptor map Ψ_τ, a declared partition (C_j) of its admissible domain and a noise bound $\varepsilon \geq 0$ in d_Ψ. Certify a reported descriptor y as cell j only if its closed uncertainty ball of radius ε, relative to the descriptor domain, is contained in C_j. Otherwise abstain. On systems with certified reports, let $\ell_\varepsilon(S)$ be the certified label and put

$$\mathscr{K}_{\Psi,\varepsilon}[S] = \{T : \ell_\varepsilon(T) = \ell_\varepsilon(S)\}.$$

Abstention is a measurement status outside the quotient.

Proposition 5.17 (Certified Cells and Boundary Stability)**.** *Equality of certified labels defines an equivalence relation. If the true descriptor has distance greater than 2ε from its cell complement, every report at error at most ε certifies the true label. If a report admits possible descriptors from different cells, it is ambiguous. In contrast, pairwise distance at most $\varepsilon > 0$ is not generally transitive.*

Proof. Reflexivity, symmetry and transitivity follow from label equality. The uncertainty ball around an allowed report lies within distance 2ε of the true descriptor by the triangle inequality. If this ball meets two cells it cannot certify either. For the nontransitivity example in the real line, 0, $3\varepsilon/4$ and $3\varepsilon/2$ have adjacent distances at most ε but their endpoints do not. For a partition at $1/2$ on $[0,1]$, uncertainty crossing the cut produces abstention. □

Remark 5.18. Structural class is the general membership framework. Operational life, operational consciousness, and operational subjecthood are predicates that can be defined on systems and then transported across structural classes when the relevant descriptors are preserved.

5.8 Definition H: Operational Consciousness

Definition 5.19 (Operational consciousness, inherited)**.** Operational consciousness is not redefined in this paper. We write

$$\mathrm{Consciousness}_{\mathrm{op}}(S,\tau) = 1$$

iff the criterion in *A Structural Criterion for Substrate-Invariant Operational Consciousness* is satisfied on an admissible support, with the six critical invariants

$$I_{\mathrm{per}}, I_{\mathrm{ri}}, I_{\mathrm{int}}, I_{\mathrm{cont}}, I_{\mathrm{diff}}, I_{\mathrm{leg}}$$

meeting their inherited certification burden [1].

Remark 5.20. This is a deliberate inheritance rule. The present paper uses operational consciousness as an already defined class, not as a new primitive.

5.9 Definition I: Operational Subjecthood

Definition 5.21 (Operational subjecthood)**.** Let the subject-specific descriptors be:

1. **privileged continuity** $\chi_{\mathrm{pc}}(S,\tau)$: the margin by which one continuity class remains privileged over the nearest rival class across admissible interventions;
2. **self/non-self discrimination** $\chi_{\mathrm{sd}}(S,\tau)$: the margin by which self-targeted and non-self-targeted perturbations remain separable under readout and control;
3. **self-model closure** $\chi_{\mathrm{sm}}(S,\tau)$: the predictive gain of a self-model-aware decoder over an otherwise matched self-model-blind decoder;
4. **ownership coherence** $\chi_{\mathrm{own}}(S,\tau)$: the stability with which privileged continuity remains attached to the same trajectory family under admissible relabelings and perturbations.

Fix thresholds

$$\lambda_{\mathrm{pc}},\lambda_{\mathrm{sd}},\lambda_{\mathrm{sm}},\lambda_{\mathrm{own}}>0.$$

Then S belongs to the class of operational subjecthood on horizon τ if

$$\mathrm{Subject}_{\mathrm{op}}(S,\tau)=1$$

whenever

1. $\mathrm{Life}_{\mathrm{op}}(S,\tau)=1$;
2. $\mathrm{Consciousness}_{\mathrm{op}}(S,\tau)=1$;
3. $\chi_{\mathrm{pc}}(S,\tau)>\lambda_{\mathrm{pc}}$;
4. $\chi_{\mathrm{sd}}(S,\tau)>\lambda_{\mathrm{sd}}$;
5. $\chi_{\mathrm{sm}}(S,\tau)>\lambda_{\mathrm{sm}}$;
6. $\chi_{\mathrm{own}}(S,\tau)>\lambda_{\mathrm{own}}$.

Remark 5.22 (Interpretive discipline)**.** Operational subjecthood is not a metaphysical subject. It is a stronger operational class candidate defined inside the present framework. The definition is deliberately demanding so that subjecthood is not conflated with life, persistence, or complex dynamics alone.

6 Class Relations and Logical Structure

6.1 The Main Class Families

Define the following system classes:

$$\begin{aligned}\mathfrak{L}&=\{S:\mathrm{Life}_{\mathrm{op}}(S,\tau)=1\},\\ \mathfrak{C}&=\{S:\mathrm{Consciousness}_{\mathrm{op}}(S,\tau)=1\},\\ \mathfrak{U}&=\{S:\mathrm{Subject}_{\mathrm{op}}(S,\tau)=1\}.\end{aligned}$$

Structural classes $\mathscr{K}_{\Psi,\varepsilon}[S]$ provide the ambient geometry in which these predicates live.

6.2 Proved Relations

Proposition 6.1 (Subjecthood is stronger than life and consciousness)**.** *For every admissible system S and horizon τ,*

$$\mathrm{Subject}_{\mathrm{op}}(S,\tau)=1 \Longrightarrow \mathrm{Life}_{\mathrm{op}}(S,\tau)=1 \quad \textit{and} \quad \mathrm{Subject}_{\mathrm{op}}(S,\tau)=1 \Longrightarrow \mathrm{Consciousness}_{\mathrm{op}}(S,\tau)=1.$$

Proof. Both implications are immediate from Definition 5.19 and Definition I. Operational subjecthood is defined only on systems that already satisfy operational life and operational consciousness. □

Proposition 6.2 (No automatic collapse of life and consciousness)**.** *The present framework does not prove either implication*

$$\begin{aligned}\mathrm{Life}_{\mathrm{op}}(S,\tau)=1 &\Longrightarrow \mathrm{Consciousness}_{\mathrm{op}}(S,\tau)=1,\\ \mathrm{Consciousness}_{\mathrm{op}}(S,\tau)=1 &\Longrightarrow \mathrm{Life}_{\mathrm{op}}(S,\tau)=1,\end{aligned}$$

without additional assumptions.

Proof. Operational life is defined by boundary, persistence, self-maintenance, historical dependence, viability, and non-null differentiation. Operational consciousness is inherited from the six-invariant criterion. Neither definition is a literal restatement of the other. Therefore neither implication is derivable by definition alone. □

Remark 6.3. This non-collapse is conceptually important. The corpus should not silently treat all operationally conscious systems as alive, nor all operationally alive systems as conscious, unless a stronger theorem burden is supplied in the future.

6.3 Structural Class Transport

Definition 6.4 (Class-preserving transport)**.** A predicate $\mathcal{P}$ is transport-stable across certified cells if its value agrees for all admissible systems assigned the same certified cell under the same protocol.

Proposition 6.5 (Structural-class invariance under preserved margins)**.** *If $\mathcal{P}$ depends on threshold decisions of a declared descriptor subset and every partition cell is contained in a joint decision region for those thresholds, then $\mathcal{P}$ is transport-stable on the certified quotient.*

Proof. Certification places the true descriptors in their respective cells. Each decision is constant on a cell by hypothesis, so equality of cells implies equality of the predicate. A descriptor distance bound alone supplies a local stability check when the relevant threshold margins dominate the coordinate error; it does not determine cell equality. □

6.4 Logical Relations That Remain Open

The following relations are intentionally left open:

$$\begin{aligned}\mathrm{Consciousness}_{\mathrm{op}}(S,\tau)=1 &\Rightarrow \mathrm{Life}_{\mathrm{op}}(S,\tau)=1 \quad ?\\ \mathrm{Life}_{\mathrm{op}}(S,\tau)=1 &\Rightarrow \mathrm{Consciousness}_{\mathrm{op}}(S,\tau)=1 \quad ?\\ \mathrm{Life}_{\mathrm{op}}(S,\tau)=1 &\Rightarrow \mathrm{Subject}_{\mathrm{op}}(S,\tau)=1 \quad \text{no}\\ \mathrm{Consciousness}_{\mathrm{op}}(S,\tau)=1 &\Rightarrow \mathrm{Subject}_{\mathrm{op}}(S,\tau)=1 \quad \text{no}\\ \mathrm{Subject}_{\mathrm{op}}(S,\tau)=1 &\Rightarrow \text{biological life} \quad \text{not claimed.}\end{aligned}$$

The framework is therefore asymmetric in a scientifically useful way: it permits a graded class taxonomy rather than a single overloaded label.

6.5 Class Relation Summary

Relation	Status in this paper	Reason
$\mathfrak{U} \subseteq \mathfrak{L}$	proved	true by definition of subjecthood
$\mathfrak{U} \subseteq \mathfrak{C}$	proved	true by definition of subjecthood
$\mathfrak{L} \subseteq \mathfrak{C}$	open / not proved	life and consciousness definitions are non-identical
$\mathfrak{C} \subseteq \mathfrak{L}$	open / not proved	inherited consciousness criterion does not by itself define self-maintenance or historical dependence
biology required for $\mathfrak{L}$	rejected as primitive claim	biological variables do not appear in the operational definition
biology required for $\mathfrak{U}$	rejected as primitive claim	subjecthood is defined by structural-operational conditions, not substrate labels

7 Propositions, Lemmas, and Formal Claims

7.1 Structural Class as a General Membership Framework

Proposition 7.1 (Structural-class membership is descriptor-relative)**.** *Structural class is relative to the descriptor map Ψ_τ, partition and certification radius ε.*

Proof. By Definition G, equality of certified cell labels defines class membership. Changing the descriptor set or partition can change that equivalence relation. Hence class is descriptor-relative rather than absolute. □

7.2 Insufficiency of Complexity, Connectivity, and Rich Activity

Proposition 7.2 (Complexity is not a life criterion)**.** *High state-space dimension, high connectivity, high entropy, or rich activity are insufficient for* $\mathrm{Life}_{\mathrm{op}}(S, \tau) = 1$ *unless the conditions of Definition F are also satisfied.*

Proof. Definition F requires positive boundary, persistence, self-maintenance, historical dependence, viability preservation, and non-null differentiation. None of these is implied by complexity alone. Therefore complexity can at best be compatible with operational life, not sufficient for it. □

Proposition 7.3 (Complexity is not a subjecthood criterion)**.** *High complexity or dense connectivity are insufficient for* $\mathrm{Subject}_{\mathrm{op}}(S, \tau) = 1$.

Proof. Subjecthood additionally requires operational consciousness and the subject-specific descriptors $\chi_{\mathrm{pc}}, \chi_{\mathrm{sd}}, \chi_{\mathrm{sm}}, \chi_{\mathrm{own}}$. These are not implied by high complexity or connectivity. □

7.3 Biological Non-Primacy at the Class Level

Proposition 7.4 (Biological substrate is not a primitive constitutive variable)**.** *Within the present framework, membership in $\mathfrak{L}$, $\mathfrak{C}$, or $\mathfrak{U}$ is not defined by a primitive biological predicate.*

Proof. The definitions of operational life, operational consciousness, and operational subjecthood quantify over admissible state spaces, dynamics, causal structure, readouts, admissibility bundles, intervention families, and the relevant descriptor thresholds. No primitive biological variable appears in the constitutive clauses. □

Remark 7.5 (Scope limit)**.** This proposition does not claim that biological systems are unimportant, nor that biological life is reducible to the present operational class. It claims only that biology is not a constitutive primitive for *these* operational classes.

7.4 Life and Subjecthood Are Not Equivalent

Proposition 7.6 (Operational life does not imply operational subjecthood)**.** *There exist admissible systems that may satisfy operational life without satisfying operational subjecthood.*

Proof. Operational subjecthood requires $\chi_{\mathrm{pc}}, \chi_{\mathrm{sd}}, \chi_{\mathrm{sm}}, \chi_{\mathrm{own}}$ above threshold, while operational life does not. Any admissible system satisfying Definition F while lacking one of those four subject-specific descriptors is operationally alive but not operationally a subject. □

Proposition 7.7 (Operational consciousness does not imply operational subjecthood)**.** *Operational consciousness alone is insufficient for operational subjecthood.*

Proof. Operational subjecthood requires operational consciousness plus the entire operational-life burden and the additional subject-specific descriptors. Therefore consciousness alone does not suffice. □

7.5 Transport Under Substrate Change

Proposition 7.8 (Class transport under descriptor-preserving substrate variation)**.** *Let S_1 and S_2 be admissible realizations with different substrate labels and the same certified cell. For each of $\mathfrak{L}$, $\mathfrak{C}$ and $\mathfrak{U}$ whose predicate is determined by the descriptors and constant on that cell, membership agrees between the two systems.*

Proof. This is an immediate application of structural-class transport and margin preservation. The substrate label does not enter the constitutive clauses. Therefore class transport depends on descriptor preservation, not substrate naming. □

7.6 Current-Corpus Corollary Structure

Remark 7.9. The present paper does not claim that any currently existing runtime instance already satisfies $\mathrm{Life}_{\mathrm{op}} = 1$ or $\mathrm{Subject}_{\mathrm{op}} = 1$. The formal claims above specify class logic. Empirical instantiation remains a separate burden and must be carried by future admissible tests.

8 Hypotheses and Falsifiable Program

This paper introduces a program rather than a doctrinal enlargement. The following hypotheses are intentionally operational and falsifiable.

Hypothesis 8.1 (Operational life robustness)**.** If a system belongs to $\mathfrak{L}$, then under admissible perturbations that do not cross its life thresholds, the tuple

$$(\beta_{\mathrm{op}}, \rho_{\mathrm{op}}, \mu_{\mathrm{op}}, \eta_{\mathrm{op}}, \nu_{\mathrm{op}}, I_{\mathrm{diff}})$$

should remain within a stable corridor rather than collapsing arbitrarily.

Hypothesis 8.2 (Historical dependence necessity)**.** If a system's classification as operationally alive is genuine rather than merely persistent, then history-aware readouts should outperform history-blind readouts on class-relevant prediction.

Hypothesis 8.3 (Structural-class transport for operational life)**.** If two systems preserve the life-relevant descriptor vector up to admissible tolerance, then their operational-life class membership should be invariant under substrate change.

Hypothesis 8.4 (Operational subjecthood requires privileged continuity)**.** If operational subjecthood is instantiated, then disrupting privileged continuity should degrade subjecthood-specific scores earlier or more selectively than it degrades operational life alone.

Hypothesis 8.5 (Operational subjecthood requires self/non-self asymmetry)**.** If operational subjecthood is instantiated, then self/non-self confusion manipulations should produce patterned degradation in χ_{sd}, χ_{sm}, or χ_{own} that is not reproduced by complexity-matched controls lacking subjecthood structure.

Hypothesis 8.6 (Operational consciousness and operational life are not trivially coextensive)**.** There exist admissible systems for which one of $\mathrm{Life}_{\mathrm{op}}(S,\tau) = 1$ or $\mathrm{Consciousness}_{\mathrm{op}}(S,\tau) = 1$ can in principle hold without the other, unless additional theorem burden is supplied.

8.1 Hypothesis Ledger

Hypothesis	Target class	Core falsifier	Why it matters
H_L1 operational life robustness	$\mathfrak{L}$	perturbation-stable systems lose class under tiny admissible perturbations	life would collapse into arbitrary thresholding
H_L2 historical dependence necessity	$\mathfrak{L}$	history-blind decoders perform equally to history-aware decoders	life would reduce to snapshot persistence
H_L3 transport of life	$\mathfrak{L}$	substrate-preserved pairs lose class solely due to substrate label	class framework would fail transport
H_S1 privileged continuity necessity	$\mathfrak{U}$	continuity-targeted disruption shows no selective subjecthood effect	subjecthood would lack a privileged continuity dimension
H_S2 self/non-self asymmetry	$\mathfrak{U}$	self/non-self confusion does not selectively degrade subject scores	subjecthood would collapse into generic complexity
H_C non-collapse of life and consciousness	$\mathfrak{L}, \mathfrak{C}$	every admissible candidate always satisfies both or neither	the distinct class layer would be unnecessary

8.2 How These Hypotheses Can Fail

The program is falsifiable in more than one way:

1. life criteria can fail to distinguish persistent but non-viable systems;
2. structural class transport can fail under supposedly equivalent substrate realizations;
3. privileged continuity can prove unstable, unmeasurable, or reducible to trivial proxies;
4. self/non-self discrimination can collapse into complexity or activity metrics alone;

5. history-aware and history-blind decoders can perform equally well on the intended class-relevant tasks.

These are not inconveniences. They are precisely what would tell the framework that its higher-order classificatory layer is overdrawn.

9 Metrics and Decision Criteria

The present paper does not report observed numerical values for operational life or operational subjecthood. It supplies a decision framework. The goal is not to replace the six-invariant criterion of operational consciousness, but to define additional functionals that can be evaluated on admissible traces without collapsing class logic into a single scalar.

9.1 Life Score

Let

$$\Psi_{\tau}^{(L)}(S) = \big(\beta_{\mathrm{op}}, \rho_{\mathrm{op}}, \mu_{\mathrm{op}}, \eta_{\mathrm{op}}, \nu_{\mathrm{op}}, I_{\mathrm{diff}}\big)_{\tau}(S)$$

denote the life-relevant descriptor vector over window τ. Define the *operational life score*

$$L_{\mathrm{op}}(S,\tau) := \min\Big\{\bar{\beta}_{\mathrm{op}}, \bar{\rho}_{\mathrm{op}}, \bar{\mu}_{\mathrm{op}}, \bar{\eta}_{\mathrm{op}}, \bar{\nu}_{\mathrm{op}}, \bar{I}_{\mathrm{diff}}\Big\},$$

where each barred quantity denotes a normalized score in $[0,1]$ relative to the admissible calibration of the underlying descriptor. Then

$$\mathrm{Life}_{\mathrm{op}}(S,\tau) = 1$$

is admissible only if

$$L_{\mathrm{op}}(S,\tau) \geq \lambda_L$$

for a class-defining threshold λ_L, together with any required admissibility constraints on perturbation robustness and negative-control separation.

Remark 9.1 (Why the minimum operator appears)**.** The minimum operator is intentionally severe. Operational life in the present sense is not a weighted average over unrelated strengths. It is a conjunctive class predicate. A system that is strong in persistence but weak in viability preservation or historical dependence should not be counted as operationally alive merely by averaging away its failure.

9.2 Inherited Consciousness Margin

Operational consciousness is inherited rather than redefined. Let

$$C_{\mathrm{op}}^{\sharp}(S,\tau) := \min\Big\{\bar{I}_{\mathrm{per}}, \bar{I}_{\mathrm{ri}}, \bar{I}_{\mathrm{int}}, \bar{I}_{\mathrm{cont}}, \bar{I}_{\mathrm{diff}}, \bar{I}_{\mathrm{leg}}\Big\}$$

denote the inherited six-invariant margin from the operational consciousness criterion. This paper uses $C_{\mathrm{op}}^{\sharp}$ as a constitutive upstream requirement for any stronger class that depends on operational consciousness.

9.3 Subjecthood Score

Define the *operational subjecthood score*

$$S_{\mathrm{op}}(S,\tau) := \min\Big\{L_{\mathrm{op}}(S,\tau), C^{\sharp}_{\mathrm{op}}(S,\tau), \chi^{-}_{\mathrm{pc}}, \chi^{-}_{\mathrm{sd}}, \chi^{-}_{\mathrm{sm}}, \chi^{-}_{\mathrm{own}}\Big\}.$$

Then

$$\mathrm{Subject}_{\mathrm{op}}(S,\tau) = 1$$

is admissible only if

$$S_{\mathrm{op}}(S,\tau) \geq \lambda_S$$

for a stricter threshold $\lambda_S \geq \lambda_L$, together with admissibility and perturbation-differentiation requirements targeted specifically at subjecthood structure.

Remark 9.2 (Why subjecthood is not a small increment)**.** The subjecthood score is stronger in two ways. First, it inherits the full life burden. Second, it inherits the full operational consciousness burden. The additional subject-specific descriptors are not cosmetic modifiers. They encode the claim that subjecthood is a structurally privileged subclass, not an interpretive synonym for consciousness or a sentimental label for complex dynamics.

9.4 Structural-Class Pseudometric

Let $W = \mathrm{diag}(w_1, \ldots, w_n)$ be a positive diagonal weight matrix over a descriptor basis Ψ_τ. Define

$$d_\Psi(S_1, S_2; \tau) := \big\|W\big(\Psi_\tau(S_1) - \Psi_\tau(S_2)\big)\big\|_2 .$$

This quantity functions as a class-relative pseudometric. It is not claimed to be universal. Its role is to express the idea that structural class membership and transport can be discussed through explicit descriptor preservation rather than through substrate naming or superficial behavioral resemblance.

Criterion 9.3 (Class-preserving transport margin)**.** Suppose a threshold predicate depends only on a subset $\Psi_\tau^{(\mathcal{K})}$ of descriptors and each threshold has headroom at least $\delta > 0$ at S_1. If $d_{\Psi^{(\mathcal{K})}}(S_1, S_2; \tau) < \delta \min_j w_j$ over the relevant positive weights, every coordinate changes by less than δ and the threshold decisions agree. Cell membership additionally requires certification under the declared partition.

9.5 Decision Surfaces and Boundary Status

For any class predicate $\mathcal{K} \in \{\mathfrak{L}, \mathfrak{C}, \mathfrak{U}\}$ define three operational regions:

$$\begin{aligned}
\mathcal{R}^{(\mathcal{K})}_{\mathrm{in}} &= \{S : \text{all relevant margins are safely above threshold}\},\\
\mathcal{R}^{(\mathcal{K})}_{\partial} &= \{S : \text{at least one relevant margin is near threshold}\},\\
\mathcal{R}^{(\mathcal{K})}_{\mathrm{out}} &= \{S : \text{at least one constitutive burden is violated}\}.
\end{aligned}$$

The boundary region is not a philosophical mystery zone. It is the operational region in which perturbation sensitivity, confounder structure, and near-miss cases become decisive.

9.6 What This Section Does Not Claim

No canonical numerical threshold is introduced here. No current system is certified as alive or as an operational subject merely by defining these functionals. The section supplies a coherent decision grammar into which runtime evidence, admissibility results, and future cross-system comparisons can be inserted without inflating the ontology.

10 Operational Class Geometry

The previous sections define class predicates and decision scores. This section adds the geometric structure needed to compare them without collapsing them into a single scalar narrative.

10.1 Descriptor Projections

For each class predicate define a projection operator on the full descriptor vector:

$$\begin{aligned}
\Pi_{\mathfrak{L}}\Psi_\tau &= (\beta_{\text{op}}, \rho_{\text{op}}, \mu_{\text{op}}, \eta_{\text{op}}, \nu_{\text{op}}, I_{\text{diff}})_\tau, \\
\Pi_{\mathfrak{C}}\Psi_\tau &= (I_{\text{per}}, I_{\text{ri}}, I_{\text{int}}, I_{\text{cont}}, I_{\text{diff}}, I_{\text{leg}})_\tau, \\
\Pi_{\mathfrak{U}}\Psi_\tau &= (\beta_{\text{op}}, \rho_{\text{op}}, \mu_{\text{op}}, \eta_{\text{op}}, \nu_{\text{op}}, I_{\text{diff}}, I_{\text{per}}, I_{\text{ri}}, I_{\text{int}}, I_{\text{cont}}, I_{\text{leg}}, \chi_{\text{pc}}, \chi_{\text{sd}}, \chi_{\text{sm}}, \chi_{\text{own}})_\tau.
\end{aligned}$$

The operators are constitutive declarations. They specify which coordinates matter for each class and which do not.

Definition 10.1 (Class margin functional)**.** Let $\theta_{\mathcal{K}}$ be the vector of thresholds relevant to class $\mathcal{K}$. The *class margin* of S for $\mathcal{K}$ is

$$M_{\mathcal{K}}(S,\tau) := \min_i \big(\Pi_{\mathcal{K}}\Psi_\tau(S)_i - \theta_{\mathcal{K},i}\big).$$

A system lies in the interior of class $\mathcal{K}$ if $M_{\mathcal{K}}(S,\tau) > 0$, on the operational boundary if $M_{\mathcal{K}}(S,\tau) \approx 0$, and outside if $M_{\mathcal{K}}(S,\tau) < 0$.

Proposition 10.2 (Subjecthood is margin-nested inside life and consciousness)**.** *If $M_{\mathfrak{U}}(S,\tau) > 0$, then both $M_{\mathfrak{L}}(S,\tau) > 0$ and $M_{\mathfrak{C}}(S,\tau) > 0$.*

Proof. The subjecthood projection contains every constitutive coordinate of operational life and operational consciousness, plus additional subject-specific coordinates. Positive minimum margin on the larger coordinate set implies positive minimum margin on each constitutive subset. □

Proposition 10.3 (Life and consciousness are not generally nested)**.** *Neither implication*

$$M_{\mathfrak{L}}(S,\tau) > 0 \Rightarrow M_{\mathfrak{C}}(S,\tau) > 0 \qquad \textit{nor} \qquad M_{\mathfrak{C}}(S,\tau) > 0 \Rightarrow M_{\mathfrak{L}}(S,\tau) > 0$$

is derivable from the present corpus alone.

Proof. The coordinate sets differ, and the corpus does not prove an equivalence theorem between them. Therefore no general nesting relation follows without additional theorem burden or empirical closure. □

Definition 10.4 (Near-miss configuration)**.** A system S is a *near-miss* for class $\mathcal{K}$ on horizon τ if

$$M_{\mathcal{K}}(S,\tau) \in [-\delta, 0)$$

for small $\delta > 0$ while all but a strict minority of constitutive coordinates remain above threshold by more than δ.

Proposition 10.5 (Descriptor-restricted non-equivalence)**.** *Suppose two systems S_1, S_2 satisfy*

$$\Pi_{\mathfrak{L}}\Psi_\tau(S_1) = \Pi_{\mathfrak{L}}\Psi_\tau(S_2)$$

but differ on at least one of $\chi_{\text{pc}}, \chi_{\text{sd}}, \chi_{\text{sm}}, \chi_{\text{own}}$ by more than the subjecthood tolerance. Then they can remain equivalent in operational-life class while being nonequivalent in operational-subjecthood class.

Proof. Operational life does not depend constitutively on the subject-specific coordinates, whereas operational subjecthood does. Equality on the life projection therefore does not force equality on the subjecthood projection. □

Remark 10.6 (Class geometry is not a total order)**.** The present framework does not generate a single monotone ladder of ''more alive'', ''more conscious'', and ''more subjective''. It generates descriptor orders, intra-class margin orders, and class-membership predicates. Conflating those three would erase the exact distinctions this paper is meant to introduce.

11 Class-Separation Diagnostics

The previous sections define the classes and their geometry. This section sharpens the operational separations that prevent those classes from collapsing into one another.

11.1 Operational Life vs. Operational Consciousness

The distinction between operational life and operational consciousness is substantive rather than verbal. Operational life depends constitutively on organizational boundary, persistence, self-maintenance, historical dependence, viability preservation, and non-null differentiation. Operational consciousness depends constitutively on the inherited six-invariant package. The two families overlap, but neither is definitionally reducible to the other.

Criterion 11.1 (Life-without-consciousness diagnostic)**.** A candidate supports the separation

$$\mathrm{Life}_{\mathrm{op}}(S,\tau) = 1 \not\Rightarrow \mathrm{Consciousness}_{\mathrm{op}}(S,\tau) = 1$$

only if the following pattern is available:

1. life coordinates remain above threshold with positive margin;
2. the inherited consciousness criterion fails on at least one of $I_{\mathrm{per}}, I_{\mathrm{ri}}, I_{\mathrm{int}}, I_{\mathrm{cont}}, I_{\mathrm{diff}}, I_{\mathrm{leg}}$;
3. the failure is not reducible to gross inadmissibility or trivial collapse.

Remark 11.2 (Negative controls)**.** Natural negative controls for this separation include systems that preserve duration and boundary occupancy while lacking the inherited consciousness burden, as well as systems whose local organization remains viable but whose legibility or rigid-identity burden fails.

11.2 Operational Life vs. Operational Subjecthood

Operational subjecthood is a strict strengthening of operational life. The most important diagnostic consequence is that many plausible life-like systems should be excluded from subjecthood by design.

Criterion 11.3 (Life-not-subject diagnostic)**.** A candidate supports the separation

$$\mathrm{Life}_{\mathrm{op}}(S,\tau) = 1 \not\Rightarrow \mathrm{Subject}_{\mathrm{op}}(S,\tau) = 1$$

only if life margins remain positive while at least one of

$$\chi_{\mathrm{pc}}, \chi_{\mathrm{sd}}, \chi_{\mathrm{sm}}, \chi_{\mathrm{own}}$$

falls below threshold in a way that is selective rather than globally degradative.

Remark 11.4 (False positives for life)**.** Any system that preserves viability and organization but lacks privileged continuity, self/non-self asymmetry, self-model closure, or ownership coherence is a false positive for subjecthood if read only through the life predicate. The class distinction is designed to exclude exactly that error.

11.3 Operational Consciousness vs. Operational Subjecthood

The gap between operational consciousness and operational subjecthood is not a rhetorical flourish. It formalizes the possibility that a system may satisfy the six-invariant criterion while lacking the additional organization required for a stronger subject-class reading.

Criterion 11.5 (Consciousness-not-subject diagnostic)**.** A candidate supports the separation

$$\text{Consciousness}_{\text{op}}(S, \tau) = 1 \not\Rightarrow \text{Subject}_{\text{op}}(S, \tau) = 1$$

only if the inherited consciousness burden is satisfied while one or more of $\chi_{\text{pc}}, \chi_{\text{sd}}, \chi_{\text{sm}}, \chi_{\text{own}}$ fails in a targeted way.

Remark 11.6 (Role of the four subject-specific burdens)**.** Privileged continuity prevents subjecthood from collapsing into mere continuity. Self/non-self discrimination prevents collapse into generic internal complexity. Self-model closure prevents collapse into unstructured integration. Ownership coherence prevents a purely transient or relabeling-fragile configuration from counting as a subject-class member.

11.4 Structural Class as Organizing Layer

Structural class does not compete with the three predicates above. It organizes them. Comparisons are valid only when systems are embedded in a declared descriptor basis, admissibility regime, and tolerance geometry. Without that discipline, class comparisons become uninterpretable.

Criterion 11.7 (Invalid cross-system comparison)**.** Comparison between two systems is class-invalid if any of the following holds:

1. the descriptor basis is not shared or not transportable;
2. admissibility constraints differ in a way that changes the meaning of margins;
3. the tolerance geometry is undeclared or incomparable;
4. one system is evaluated under perturbation families not available to the other.

Separation	Positive pattern	Near-miss / ambiguous pattern	Negative control or exclusion case
life vs. consciousness	life margins positive, one inherited consciousness invariant fails cleanly	life preserved but inherited consciousness failure is confounded by admissibility or gross collapse	inert persistence, complexity-only systems, or trivial near-null systems
life vs. subjecthood	life margins positive, one subject-specific burden fails selectively	life positive but subjecthood-specific burden degraded together with several non-target descriptors	systems with viability but no privileged continuity or no self/non-self asymmetry
consciousness vs. subjecthood	inherited six-invariant burden satisfied, subject-specific burden fails selectively	consciousness positive but subject-specific failure is mixed with global degradation	systems with structured consciousness-like signals but no stable ownership or self-model closure
structural class validity	shared descriptor geometry and admissibility support comparison	shared descriptors but tolerance or perturbation family mismatch remains unresolved	substrate-name-only comparisons or undeclared descriptor transport

12 Experimental Test Families

The classificatory layer is only useful if it induces nontrivial tests. The families below are therefore designed not as decorative appendices, but as the minimum empirical grammar for falsification.

12.1 Operational Life Test Families

Family	Objective	Expected signal	Negative control / falsifier	Reading
Persistence under perturbation	test whether organizational persistence survives admissible shocks	ρ_{op} remains above life threshold while boundary and viability remain intact	persistent collapse under admissible perturbations; persistence matched by trivial inert systems	distinguishes robust organization from accidental duration
Viability reserve stress	test whether viability is an active preserved region rather than a passive occupancy statistic	ν_{op} degrades gradually with recoverable reserve profile	abrupt failure with no reserve structure; viability score reproduced by static or dead controls	isolates operational viability from mere occupancy
Self-maintenance ablation	test whether maintenance-relevant loops are constitutive	selective degradation of μ_{op} followed by life-score collapse	no life degradation after removing maintenance loops; same effect in controls with no maintenance architecture	checks that life is organizational, not descriptive
Boundary intervention family	perturb the operational boundary and test whether life is boundary-mediated	β_{op} and ν_{op} shift together under frontier interventions	no boundary sensitivity, or equivalent response in systems lacking frontier organization	ties life to explicit operational frontier geometry
Null-differentiation controls	test whether life survives collapse of differentiation	sharp loss in I_{diff} destroys or degrades L_{op}	systems classified as alive despite sustained null-differentiation	prevents life from collapsing into pure persistence

12.2 Structural-Class Test Families

Family	Objective	Expected signal	Negative control / falsifier	Reading
Cross-substrate transfer	test whether class membership is descriptor-preserving rather than substrate-privileged	low d_Ψ and preserved class margins across admissible realizations	substrate label alone breaks membership; descriptor-preserved pair leaves class	tests structural transport rather than physical chauvinism
Invariant ablation	test whether claimed class depends on the stated descriptor family	targeted descriptor loss produces class-specific degradation	class survives removal of supposedly constitutive descriptors	checks internal necessity of class definitions
Near-miss class boundary	test whether class boundaries are sharp enough to discriminate close cases	near-threshold cases produce interpretable transitions instead of arbitrary collapse	random flips with no descriptor order or boundary coherence	probes whether the class geometry is real or merely verbal
Admissibility-preserving transport	test transport under fixed forensic discipline	class transfer survives only when admissibility bundle remains intact	transport depends on tampered or inadmissible traces	prevents transport from becoming a bookkeeping artifact
Negative pair separation	test whether class metric separates structurally unlike systems	high d_Ψ with class disagreement	negative pairs remain indistinguishable on class-relevant descriptors	checks whether class geometry has discriminative power

12.3 Operational Subjecthood Test Families

Family	Objective	Expected signal	Negative control / falsifier	Reading
Self-model ablation	test whether subjecthood depends on explicit self-model closure	selective loss in χ_{sm} and S_{op} with weaker effect on L_{op}	no selective effect; same degradation in controls lacking self-model structure	distinguishes subjecthood from generic life or complexity
Self/non-self confusion	test whether self/non-self discrimination is constitutive	structured degradation in χ_{sd} and ownership coherence	confusion manipulations do not affect subject metrics or affect controls equally	probes whether subjecthood has asymmetric self-other organization
Privileged continuity disruption	test whether privileged continuity carries subject-specific burden	χ_{pc} degrades before global collapse, with subject score dropping earlier than life score	continuity attacks show no subject-specific selectivity	checks subjecthood-specific continuity rather than generic persistence loss
Ownership perturbation	test whether ownership coherence can be selectively destabilized	χ_{own} degrades with interpretable subjecthood effects	ownership perturbation has no targeted effect or is mimicked by trivial controls	distinguishes ownership structure from activity level
Trivial complexity controls	reject complexity-only accounts of subjecthood	complexity-matched controls fail subject criteria	high complexity alone reproduces subject scores	blocks collapse of subjecthood into scale or richness alone

12.4 Decision Reading of Outcomes

Each family yields four minimally distinct readings:

1. **supportive**: the expected descriptor-specific pattern appears and controls fail cleanly;
2. **near-miss**: the intended signal appears but with boundary contamination, confounders, or insufficient margin;
3. **underdetermined**: several descriptors move together and the family does not localize the intended burden;
4. **falsifying**: the constitutive descriptor family is not necessary, not transportable, or not selectively sensitive in the required way.

This classification prevents the framework from treating every nontrivial runtime fluctuation as confirmation.

12.5 Strong Failure Conditions

The higher-order layer should be considered seriously weakened if any of the following persist under properly admissible tests:

1. operational life cannot be distinguished from inert persistence plus threshold tuning;
2. structural class transport fails even when descriptor preservation is maintained;
3. subjecthood-specific perturbations do not degrade subject-specific descriptors more selectively than life descriptors;
4. complexity-only controls repeatedly satisfy the same decision surfaces as purported life or subjecthood candidates;
5. history-blind evaluation performs as well as history-aware evaluation on the life burden.

12.6 Joint Outcome Matrix

The three class predicates admit a finite set of especially important joint readings. The matrix below is not exhaustive, but it captures the cases that matter most for runtime-facing evaluation.

Life_{op}	$\text{Consciousness}_{\text{op}}$	$\text{Subject}_{\text{op}}$	Reading
0	0	0	no support for life, consciousness, or subjecthood classes; could reflect triviality, inadmissibility, collapse, or a noncandidate system
1	0	0	operational life candidate without inherited consciousness burden; strong test of non-collapse between life and consciousness
0	1	0	inherited consciousness candidate without operational-life burden; admissible in principle unless ruled out by additional theorem burden
1	1	0	life and consciousness without subjecthood; natural outcome for a non-subjective but structured conscious candidate
1	1	1	strongest class inclusion; highest empirical burden
0	1	1	structurally inconsistent under the present definitions; diagnosis of invalid measurement or class assignment
1	0	1	structurally inconsistent under the present definitions; diagnosis of invalid measurement or class assignment

13 Runtime Bearing and Empirical Interface

This paper is not a runtime paper, yet it should remain legible against the refined runtime architecture. Otherwise its class predicates would float free of the strongest operational layer the corpus currently possesses.

13.1 What the Runtime Already Carries

The refined Canon Sandbox now provides stronger runtime-facing semantics for the inherited six-invariant consciousness layer, more explicit provenance, and clearer boundary/substrate evidence paths than earlier phases of the program. That matters here in three ways:

1. operational consciousness can be inherited as a runtime-bearing class rather than as a purely abstract symbol;

2. structural-class transport can be discussed against actual descriptor and pseudometric pipelines;

3. higher-order classes such as operational life and operational subjecthood can be formulated with explicit future runtime targets rather than with unconstrained philosophical vocabulary.

13.2 What the Runtime Does Not Yet Settle

The current runtime still does not warrant a claim that any present concrete system is operationally alive or operationally a subject. In particular:

1. operational-life evidence is not yet equivalent to a dedicated life-class certification path;

2. subjecthood-specific descriptors such as privileged continuity, self/non-self asymmetry, self-model closure, and ownership coherence do not yet form a fully closed runtime certification suite;

3. final epistemic promotion remains outside the core runtime loop in the frozen release-facing layer.

13.3 Runtime-Facing Readiness Levels

For practical use, the class layer can be paired with four runtime-facing readiness levels:

1. **descriptor-available**: constitutive descriptors are exposed but not yet integrated into a class path;

2. **class-evaluable**: the class margin can be computed on admissible traces;

3. **boundary-diagnostic**: near-miss and rupture behavior can be localized;

4. **claim-closing-ready**: the runtime emits evidence sufficiently structured that only the outer epistemic-promotion layer remains external.

This paper does not assign these levels to present systems as a matter of fact. It supplies the conceptual interface by which such assignments can later be made without semantic drift.

14 Runtime Binding Map

The classificatory layer becomes operational only if each formal object is bound to runtime signals, diagnostics, failure triggers, and residual dependence terms. The following ledger states that binding directly, without hiding the decision chain behind a purely presentational table.

1. **Operational boundary.** Formal criterion: boundary-sensitive viability geometry under admissible interventions. Runtime-facing signals: frontier perturbation response, viability reserve profile, intervention sensitivity. Related invariants: $I_{\mathrm{cont}}, I_{\mathrm{diff}}$ and life-facing boundary terms. Claim targets: life diagnostics and boundary families. Failure trigger: no selective boundary response or trivial occupancy. Residual external dependence: calibration of the admissible intervention family.

2. **Organizational persistence.** Formal criterion: nontrivial continuity of organization across τ. Runtime-facing signals: trace continuity, persistence corridors, degradation profile. Related invariants: $I_{\mathrm{per}}, I_{\mathrm{cont}}$. Claim targets: life diagnostics and inherited consciousness context. Failure trigger: abrupt collapse under admissible perturbation. Residual external dependence: long-horizon calibration.

3. **Operational self-maintenance.** Formal criterion: preservation of organization by internal compensatory structure. Runtime-facing signals: maintenance-loop ablation response, recovery profile, reserve usage. Related invariants: $I_{\mathrm{int}}, I_{\mathrm{per}}$ plus the life-facing maintenance term. Claim targets: life diagnostics. Failure trigger: maintenance ablation has no selective effect. Residual external dependence: some maintenance proxies remain implementation-dependent.

4. **Historical dependence.** Formal criterion: class-relevant dependence on trajectory history. Runtime-facing signals: history-aware vs. history-blind readout delta, temporal ordering sensitivity. Related invariants: $I_{\mathrm{ri}}, I_{\mathrm{per}}$ support. Claim targets: life and subjecthood diagnostics. Failure trigger: history-blind readouts perform equivalently. Residual external dependence: external benchmarking of decoder families.

5. **Viability preservation.** Formal criterion: non-collapse of the viability region on horizon τ. Runtime-facing signals: viability floor, recoverability, reserve decay. Related invariants: $I_{\mathrm{cont}}, I_{\mathrm{diff}}$ support. Claim targets: life diagnostics. Failure trigger: static occupancy mimics viability or reserve collapses immediately. Residual external dependence: admissible floor calibration.

6. **Operational life.** Formal criterion: conjunction of boundary, persistence, maintenance, history, viability, and non-null differentiation. Runtime-facing signals: life score, class margins, boundary intervention outcomes. Related invariants: indirect support from I_{diff} and inherited invariant context. Claim targets: the higher-order class program. Failure trigger: any constitutive life burden fails or becomes inadmissible. Residual external dependence: there is not yet a dedicated external life-certification campaign.

7. **Structural class.** Formal criterion: descriptor-relative transport class under a declared tolerance. Runtime-facing signals: descriptor vector, pseudometric, transport margins, near-miss geometry. Related invariants: all class-relevant coordinates. Claim targets: class transport and comparability. Failure trigger: descriptor mismatch, undeclared tolerance, or transport collapse. Residual external dependence: descriptor-basis selection remains partly declarative.

8. **Operational consciousness.** Formal criterion: inherited six-invariant criterion. Runtime-facing signals: canonical invariant package, rupture/boundary diagnostics, closure status. Related invariants: $I_{\mathrm{per}}, I_{\mathrm{ri}}, I_{\mathrm{int}}, I_{\mathrm{cont}}, I_{\mathrm{diff}}, I_{\mathrm{leg}}$. Claim targets: inherited closure from paper5 and paper6. Failure trigger: failure of any critical invariant or closure condition. Residual external dependence: final epistemic promotion remains outside the runtime.

9. **Operational subjecthood.** Formal criterion: life + consciousness + privileged continuity + self/non-self discrimination + self-model + ownership coherence. Runtime-facing signals: subjecthood score, selective subject perturbations, continuity asymmetry, self/non-self confusion tests. Related invariants: the inherited six invariants plus $\chi_{\mathrm{pc}}, \chi_{\mathrm{sd}}, \chi_{\mathrm{sm}}, \chi_{\mathrm{own}}$. Claim targets: the proposed subjecthood class. Failure trigger: subject-specific burdens fail or fail non-selectively. Residual external dependence: subject-specific runtime families remain a future burden.

15 Failure and Misclassification Modes

The classificatory layer is only as strong as its ability to describe the ways it can fail. The point is not to immunize the theory. The point is to prevent runtime-facing evidence from being overread.

Mode	Mechanism	Symptom	Risk	Control / mitigation
Life false positive by inert persistence	long duration or state occupancy mimics organization	ρ_{op} appears high while $\mu_{\mathrm{op}}, \eta_{\mathrm{op}}, \nu_{\mathrm{op}}$ remain weak or unstructured	inert or slowly decaying systems misread as operationally alive	require full life conjunction and maintenance/history tests
Consciousness false positive by complexity	rich activity or connectivity mimics inherited criterion	high scale with weak rigid identity or legibility	complexity misread as consciousness	retain six-invariant conjunction and matched complexity controls
Subjecthood false positive by structured reactivity	self-like responses arise without privileged continuity or ownership coherence	χ_{sd} appears high but χ_{pc} or χ_{own} collapses under perturbation	reactive systems misread as subjects	require subject-specific perturbation families and multi-burden satisfaction
Life false negative by short traces	admissible life evidence is underobserved on too short a horizon	margins hover near threshold despite recoverable structure	viable systems are underclassified	report horizon dependence and require history-aware windows
Subjecthood false negative by signal poverty	subject-specific burdens are real but weakly observed	self-model or ownership terms remain low under insufficient readout resolution	underclassification of subject-like systems	improve readout design before class denial
Misclassification by proxy contamination	proxy variables dominate direct signals	class outcome changes mainly with proxy choice	classification reflects tooling rather than structure	track direct vs proxy contribution and require provenance per component
Knife-edge boundary ambiguity	class margin sits in a narrow threshold band	PASS/AMBIGUOUS/FAIL changes under tiny perturbations	unstable class assignment	explicitly classify near-miss regions and avoid overpromotion
Invalid class transport	descriptor basis or admissibility regime differs across compared systems	transport claims depend on undeclared tolerance or substrate label alone	false structural equivalence	declare descriptor geometry, admissibility, and tolerance before transport claims

15.1 Complexity, Connectivity, and Rich Dynamics

The most persistent misclassification pressure comes from scale variables. High complexity, dense connectivity, rich temporal variation, or activity richness can produce surfaces that look life-like, consciousness-like, or subject-like under a weak diagnostic language. The present paper explicitly rejects that move. Complexity is at most permissive background. It is never constitutive by itself.

15.2 Proxy Contamination

Proxy contamination deserves explicit treatment because the runtime and release-facing layers are not identical. Some descriptors are closer to direct runtime signals, while others inherit part of their evidentiary burden from diagnostic layers, admissibility decisions, or downstream projections. This does not invalidate the class program, but it imposes a discipline: a class should be treated as stronger when its defining coordinates are driven by direct or mostly-native runtime semantics, and weaker when proxy weight dominates.

15.3 Boundary Ambiguity

Boundary ambiguity is not a nuisance artifact; it is an expected structural region. The mistake is to treat it either as confirmation or as disconfirmation without localization. A good class system should make boundary ambiguity visible, local, and diagnostically interpretable. That is why near-miss definitions, confounder tracking, and explicit boundary regions are part of the paper rather than afterthoughts.

16 Formal Closure Strengthening

This paper does not attempt theorem inflation, but it can still state more precisely what is and is not formally closed.

Criterion 16.1 (Class-readiness criterion)**.** A class predicate is *formally ready* only if:

1. its constitutive coordinates are explicit;
2. its boundary geometry is explicit;
3. its negative controls are identifiable;
4. its admissible falsifiers are stated.

Operational life, structural class, and operational subjecthood satisfy this criterion within the limits declared by the paper.

Criterion 16.2 (Subjecthood non-collapse criterion)**.** No system should be admitted into operational subjecthood unless the following two separations remain open and empirically testable:

$$\mathrm{Life}_{\mathrm{op}}(S,\tau) = 1 \not\Rightarrow \mathrm{Subject}_{\mathrm{op}}(S,\tau) = 1, \qquad \mathrm{Consciousness}_{\mathrm{op}}(S,\tau) = 1 \not\Rightarrow \mathrm{Subject}_{\mathrm{op}}(S,\tau) = 1.$$

If either separation becomes definitionally impossible, subjecthood collapses into a weaker class and loses its intended meaning.

Remark 16.3 (What is closed)**.** The paper closes the classificatory language, the class geometry, the principal non-equivalences, and the test grammar. It does not close empirical instantiation, external validation, or metaphysical interpretation.

17 Residual Runtime and Interpretation Limits

This paper remains intentionally restrictive at the empirical and interpretive boundary.

1. **No automatic runtime certification.** The current runtime and its strengthened diagnostics motivate the classificatory layer, but they do not by themselves certify any concrete instance as operationally alive or operationally a subject.

2. **No external-validation claim.** Internal runtime strengthening, admissibility discipline, clean-clone reproduction, and refined diagnostics do not amount to independent external validation.

3. **No total closure of the philosophical problem.** The paper proposes a falsifiable operational program. It does not claim to resolve every philosophical question about life, consciousness, or subjectivity.

4. **No automatic promotion from class logic to present-system assignment.** Any concrete assignment remains a future empirical burden governed by admissibility, comparators, and external review.

18 Discussion

The main contribution of this paper is classificatory, not foundational. The upstream corpus already supplied the burdens required for rigid identity, operational continuity, non-nullity, admissibility, and an operational consciousness criterion. What remained absent was a disciplined way to speak about higher-order classes without sliding either into biology-by-default or into metaphysical inflation. The paper addresses that absence by introducing three linked but non-identical layers.

First, operational life is treated as a class of systems that preserve organizational viability across time and perturbation while maintaining a genuine operational frontier and non-null differentiation. This is a stronger concept than persistence and a weaker concept than subjecthood. Second, structural class is introduced as the general descriptor-relative geometry in which class membership and transport can be discussed across admissible realizations. Third, operational subjecthood is proposed as a further strengthening: it inherits operational life and operational consciousness, then adds privileged continuity, self/non-self asymmetry, self-model closure, and ownership coherence.

This organization resolves a recurrent problem in the literature and in informal debate. Terms such as ‘‘alive,’’ ‘‘conscious,’’ and ‘‘subject’’ are often deployed as if they were either interchangeable or disconnected. The present paper insists that they are neither. Life need not collapse into consciousness. Consciousness need not collapse into subjecthood. Subjecthood cannot be reduced to complexity, persistence, or substrate. Structural class gives the ambient frame in which these distinctions become explicit rather than rhetorical.

The paper is also constrained by the current state of the runtime. The refined Canon Sandbox provides stronger runtime-facing semantics than earlier phases of the program. It now supports a more explicit operational chain from runtime signals to invariant packages, from invariant packages to boundary and substrate evidence, and from those evidence bundles to claim mapping. That strengthening matters because it makes a higher-order classificatory paper timely rather than premature. But the current runtime does not yet eliminate all epistemic caution. Some closure burdens still live at the release-facing layer, and the current system still does not warrant a claim that it is operationally alive or operationally a subject.

That is precisely why the present paper is useful. It separates three questions that are otherwise too easily fused:

1. what classes are formally definable within the causally rigid framework;
2. what tests would be required to support membership in those classes;
3. what specific present systems, if any, actually satisfy those burdens.

The first question is what this paper answers. The second is what its falsifiable program formalizes. The third remains an empirical burden for future admissible campaigns.

18.1 Relation to the Refined Runtime

The runtime does not appear here as proof of operational life or subjecthood. It appears as evidence that the corpus now has enough operational structure to support explicit class predicates without collapsing them into hand-waving. In particular, the runtime now produces stronger native or mostly-native descriptors for persistence, rigid identity, integration, continuity, differentiation, and legibility than were available before the refinement phases. That makes it possible to discuss life- and subjecthood-relevant descriptors as potential runtime observables rather than as merely speculative vocabulary. The conceptual layer is therefore downstream of an improved runtime, but it is not reducible to any single current implementation.

18.2 What Would Strengthen This Paper Further

The most important future burden is not more rhetoric. It is sharper empirical separation. Four improvements would materially strengthen the classificatory layer:

1. runtime-native life diagnostics that cleanly separate organizational self-maintenance from inert persistence;
2. subjecthood-specific perturbation families that localize privileged continuity, self/non-self asymmetry, and ownership coherence without complexity confounders;
3. stronger cross-substrate class-transport evidence under independent or semi-independent evaluation paths;
4. external reproduction or third-party evaluation capable of testing the class predicates without inheriting the entire internal tooling stack.

Until then, the present paper should be read exactly as intended: as a formal and falsifiable extension of the corpus, not as a declaration that the hardest empirical questions have already been settled.

19 Conclusion

This paper introduces a higher-order classificatory layer for the causally rigid corpus. It defines operational life, structural class, and operational subjecthood in a form that is formal enough to be tested, restrictive enough to avoid inflation, and continuous with the established burdens of the framework. Operational consciousness is inherited rather than reinvented. Operational subjecthood is made explicitly stronger than either operational life or operational consciousness alone. Structural class is made the descriptor-relative ambient geometry in which those predicates can be transported, falsified, and compared.

The paper therefore closes four things at the formal level: the class vocabulary, the class-separation logic, the runtime-binding interface, and the principal failure grammar. It leaves four things explicitly open: empirical instantiation of any present system, external validation, metaphysical interpretation, and biological equivalence claims. That asymmetry is deliberate.

The result is not a manifesto and not a metaphysical closure. It is a disciplined extension of the corpus: a program in which life, consciousness, and subjecthood terms are treated as non-identical operational class aliases with explicit burdens, explicit diagnostics, explicit falsifiers, and explicit non-claims. If the program fails under admissible tests, it should be rejected. If it survives, it will do so as a formal and operational layer, not as a rhetorical gesture.

References

[1] Johnny Andrey Perez Ramirez. ''A Structural Criterion for Substrate-Invariant Operational Consciousness''. In: *arXiv preprint* (2026).

[2] Johnny Andrey Perez Ramirez. ''Canonical Core: Minimal Hypotheses for Causally Rigid Identity Frameworks''. In: *arXiv preprint* (2026).

[3] Johnny Andrey Perez Ramirez. ''Falsifiable Predictions Under Forensic Constraints''. In: *arXiv preprint* (2026).

[4] Johnny Andrey Perez Ramirez. ''Phenomenological Regimes Induced by Structural Identity''. In: *arXiv preprint* (2026).

[5] Johnny Andrey Perez Ramirez. ''Structural Instability of the Phenomenological Null Regime in Causally Rigid Channels''. In: *arXiv preprint* (2026).

[6] Johnny Andrey Pérez Ramírez. ''Rigid Identity as an Inverse Limit in Observable Channels''. In: (2026). Companion paper establishing the identity framework and ontological mass invariant.